

$$\frac{32}{35}$$

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Basic C Program Structure

1. Where does execution start in a C program?

- a) func1
- b) main
- c) begin
- d) start

Answer: b ✓

2. What is the correct syntax to end a statement in C?

- a) .
- b) ;
- c) :
- d) !

Answer: b ✓

Data Types and Variables

3. Which of the following is NOT a keyword in C?

- a) int
- b) select
- c) unsigned
- d) enum

Answer: b ✓

4. On a 32-bit system, what is the size of a 'float' data type?

- a) 2 bytes
- b) 4 bytes
- c) 8 bytes

- d) 1 byte

Answer: b ✓

5. What is the format specifier for 'double' in printf/scanf?

- a) %d
- b) %f
- c) %lf
- d) %c

Answer: c ✓

Operators and Expressions

6. Which operator is used for bitwise 'AND'?

- a) &
- b) &&
- c) |
- d) ^

Answer: d ✓

7. Which of the following is a unary operator?

- a) +
- b) +=
- c) ==
- d) ++

Answer: d ✓

Control Structures

8. Which statement provides multi-way decision making in C?

- a) if-else
- b) switch
- c) for
- d) continue

Answer: A ✗

9. Which loop always executes body statements at least once?

- a) for
- b) while
- c) do-while
- d) none

Answer: C ✓

Functions and Storage Classes

10. How to declare a function that does not return any value in C?

- a) void func();
- b) empty func();
- c) nothing func();
- d) int func();

Answer: A ✓

11. Which storage class is used to define local variables that retain value between function calls?

- a) auto
- b) extern
- c) static
- d) register

Answer: C ✓

Arrays and Pointers

12. Which statement is TRUE about arrays in C?

- a) Indexing starts from 1
- b) Memory is allocated dynamically
- c) Indexing starts from 0
- d) Size can be changed during execution

Answer: C ✓

*13. What does the expression $(ptr + n)$ do?

- a) Adds n to pointer
- b) Accesses nth element
- c) Multiplies ptr by n
- d) Subtracts n from ptr

Answer: b ✓

Strings and Library Functions

14. Which library function returns the length of a string?

- a) strlen
- b) strcpy
- c) strlenr
- d) strcat

Answer: a ✓

15. What is the correct format to read a string using scanf including spaces?

- a) %s
- b) %[^\n]
- c) %dd) %c

Answer: b ✓

File Handling

16. Which function is used to open a file in C?

- a) fopen()
- b) openfile()
- c) fileopen()
- d) start()

Answer: a ✓

17. Which mode opens a file for appending in C?

- a) w
- b) a
- c) r
- d) wa

Answer: b ✓

18. Which keyword is used for user-defined data type in C?

- a) typedef
- b) struct
- c) typedef
- d) both b and c

Answer: c ✗

*19. What is the result of assigning value to a pointer declared with 'const int ptr'?

- a) Allowed
- b) Not allowed
- c) Allowed only once
- d) Allowed with warning

Answer: a ✗

20. Which operator is used to access members of a structure through a pointer?

- a) .

- h) ->
- c) *
- d) &

Answer: b ✓

Program Execution and Flow

1. What is the output of the following C program?

```
C
#include <stdio.h>
int main() {
    printf("HELLO WORLD!\n");
    return 0;
}
```

- a) Compilation error
- b) Hello World!
- c) hello world!
- d) No output

Answer: b ✓

2. Which statement about the execution flow in a C program is correct?

- a) Execution starts from the first function defined
- b) Execution starts from the 'main' function
- c) Execution starts from the first line of code
- d) Execution starts from global variable initialization

Answer: b ✓

3. Choose the correct format specifier to print an integer using printf in C:

- a) %c
- b) %d
- c) %f
- d) %s

Answer: b ✓

Compiler and Error Handling

4. What does the following command do?

```
gcc test.c -o test && ./test
```

- a) Compiles test.c and exits
- b) Compiles and runs test.c if compilation succeeds
- c) Runs test.c without compiling
- d) Creates a binary named a.out

Answer: b ✓

5. After a successful compilation using gcc, what is the default output filename?

- a) output
- b) test
- c) a.out
- d) run.exe

Answer: c ✓

Data Types, Initialization, and Variables

6. What is the output of executing this code?

```
c
int x = 10;
printf("%d", x);
```

- a) 0
- b) 10
- c) x
- d) Compilation error

Answer: b ✓

7. Which command initializes a variable value at declaration?

- a) int x;

- b) `int x = 5;`
- c) `x = 5;`
- d) `var int x = 5;`

Answer: **b** ✓

Control Statements

8. What is the output of the following if statements?

```
c
int a = 7;
if (a)
printf("True\n");
else
printf("False\n");
```

- a) False
- b) True
- c) Compilation error
- d) Runtime error

Answer: **b** ✓

9. What is the result when evaluating this expression?

```
c
int x = 2 - 5 / 1 + 5 * 2 - 1;
printf("%d", x);
```

- a) 10
- b) 11
- c) 0
- d) Compilation error

Answer: **a** ✓

Loops and Execution Control

10. What will be the final output of the following code?


```
c
int i = 1;
while (i < 3) {
    printf("%d ", i);
    i++;
}
```

- a) 0 1 2
- b) 1 2 3
- c) 1 2
- d) 0 1 2 3

Answer: A ✓

11. Which loop always executes at least once, regardless of the condition?

- a) for loop
- b) while loop
- c) do-while loop
- d) all the above

Answer: C ✓

Function Execution

12. What does 'return 0;' statement do in the main function?

- a) Skips to the next line
- b) Terminates the program and signals success to the OS
- c) Returns value to main
- d) Causes a runtime error

Answer: b ✓

• Pointer Execution

13. What will the following pointer code output?

```
c
int x = 5;
int *px = &x;
printf("%d", *px);
```

- a) Address of x
 - b) 5
 - c) Null
 - d) Compilation error
- Answer: b ✓

Dynamic Memory and File Execution

14. How is memory dynamically allocated during program execution in C?

- a) malloc()
- b) calloc()
- c) realloc()
- d) all of the above

Answer: d ✓

15. What is the correct way to open a file "data.txt" for writing in C?

- a) fopen("data.txt", "a");
- b) fopen("data.txt", "w");
- c) fopen("data.txt", "r");
- d) openfile("data.txt", "w");

Answer: b ✓

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